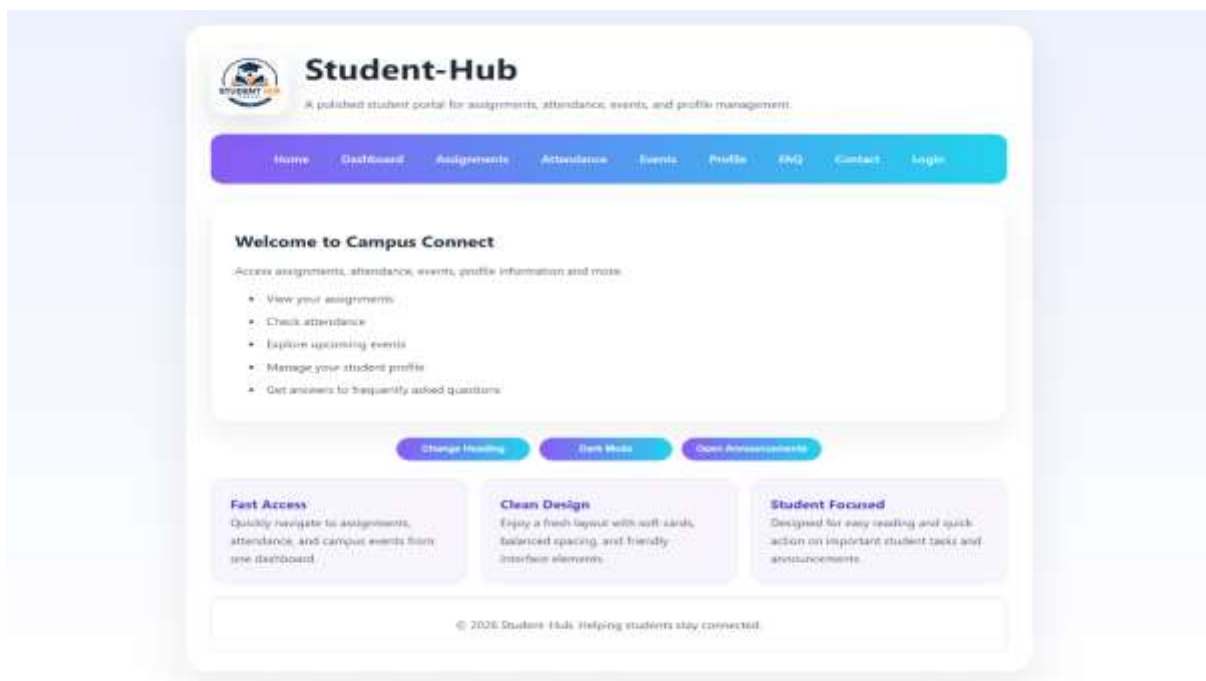


PRACTICAL – 6

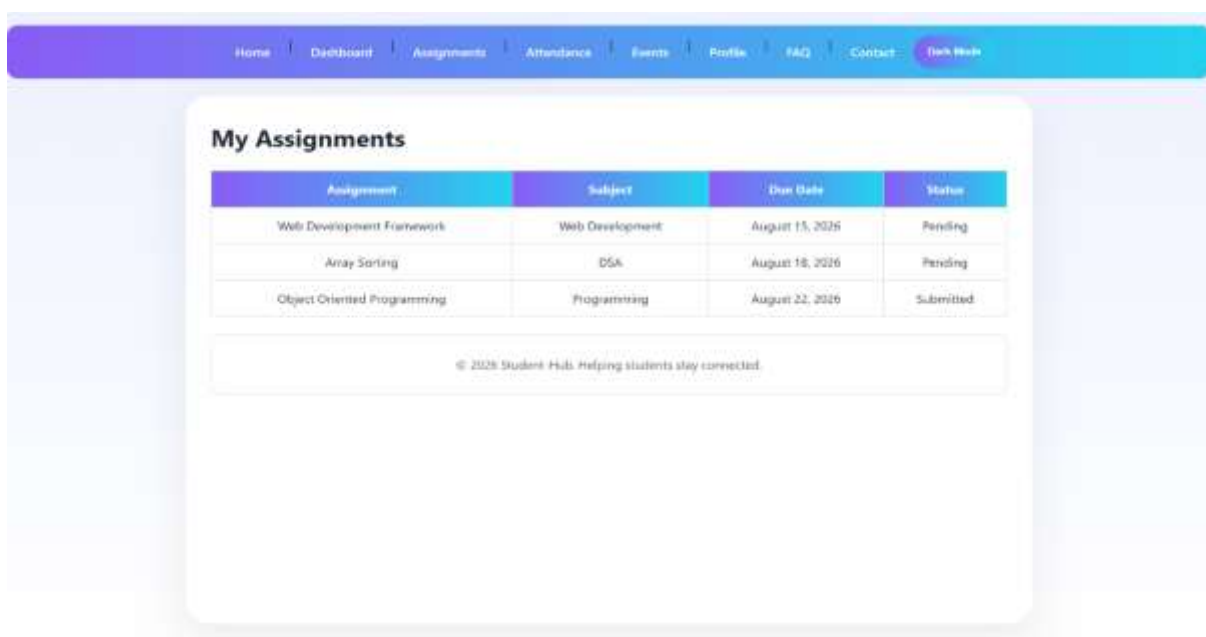
Problem Definition: Rendering External JSON Data using Fetch API, Search and Filter
 Problem Definition Display event lists, student profiles, notices, or FAQs from external JSON files using Fetch API. Implement dynamic rendering, search, filter, sorting, and pagination.

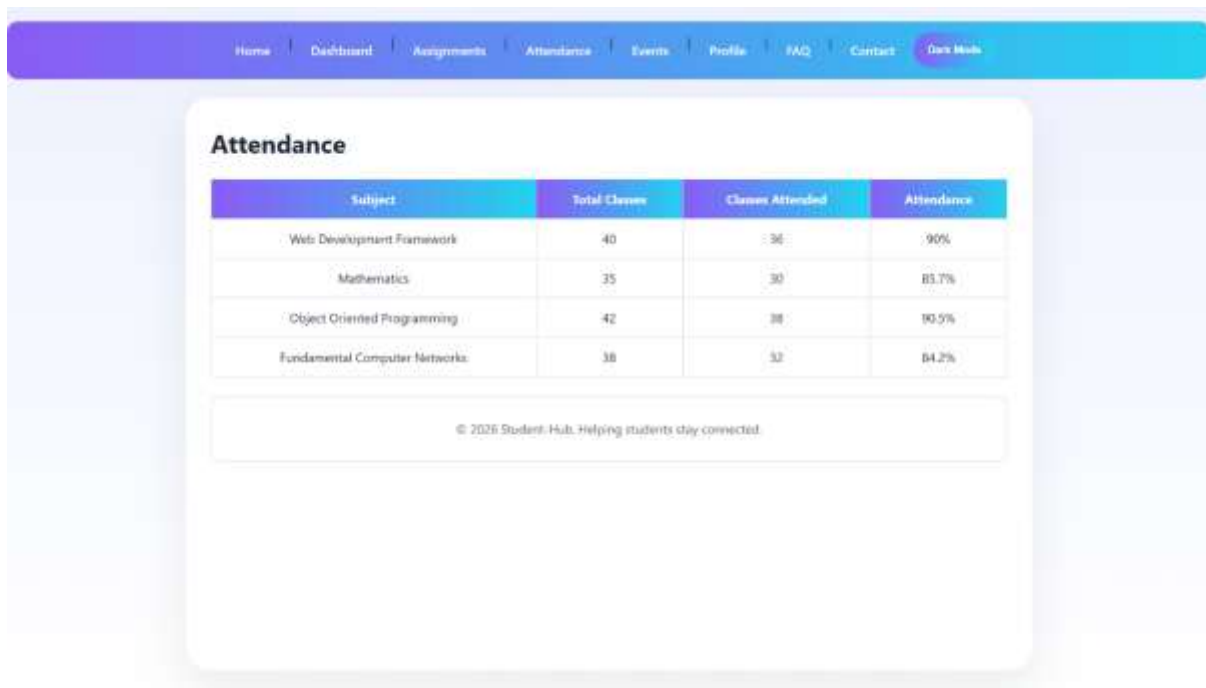
Output:

Home Page:



Assignment Page:

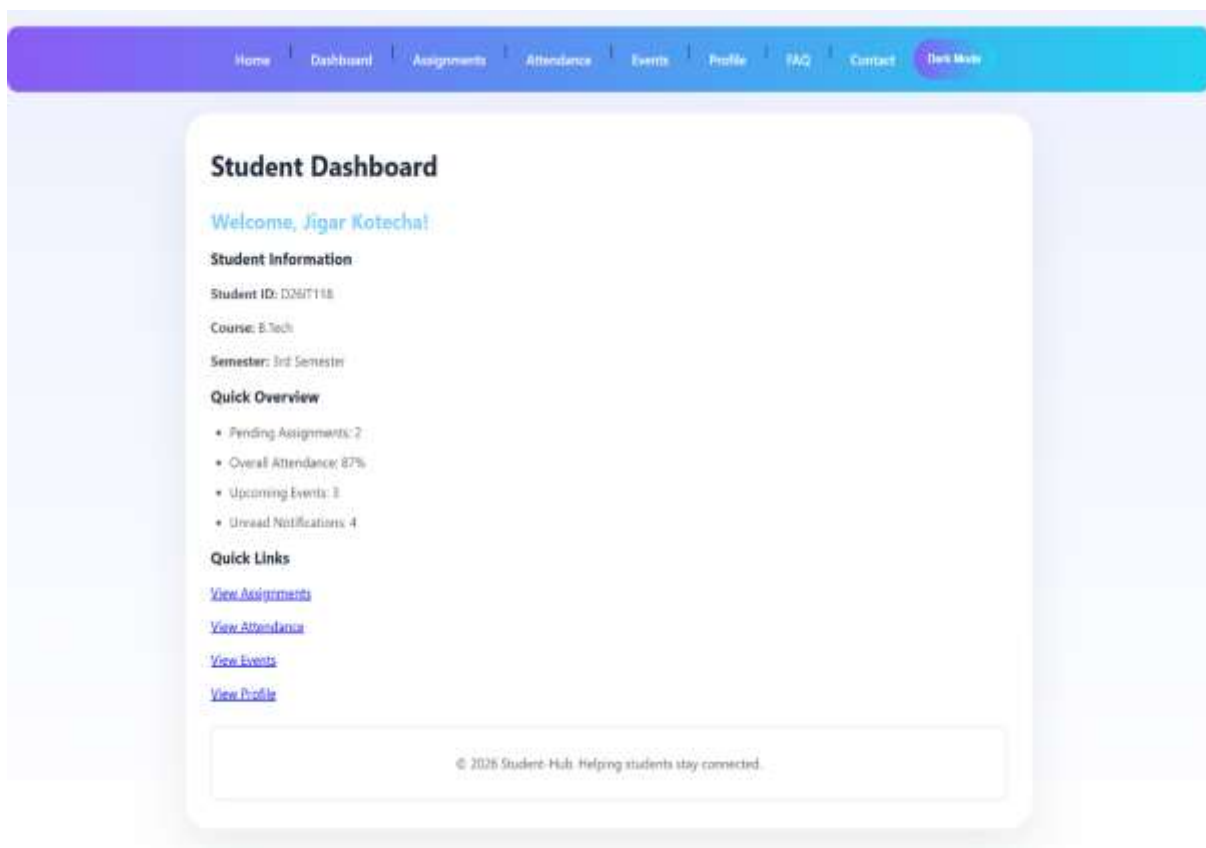


Attendance Page :


The screenshot shows the 'Attendance' page of a student portal. It features a navigation bar with links: Home, Dashboard, Assignments, Attendance, Events, Profile, FAQ, Contact, and a Dark Mode toggle. The main content area is titled 'Attendance' and contains a table with the following data:

Subject	Total Classes	Classes Attended	Attendance
Web Development Framework	40	36	90%
Mathematics	35	30	85.7%
Object Oriented Programming	42	38	90.5%
Fundamental Computer Networks	38	32	84.2%

At the bottom of the page, there is a footer: © 2026 Student Hub. Helping students stay connected.

dashboard page :


The screenshot shows the 'Student Dashboard' page of a student portal. It features the same navigation bar as the Attendance page. The main content area is titled 'Student Dashboard' and includes a welcome message: 'Welcome, Jigar Kotecha!'. Below this, there is a 'Student Information' section with the following details:

- Student ID: D26IT118
- Course: B.Tech
- Semester: 3rd Semester

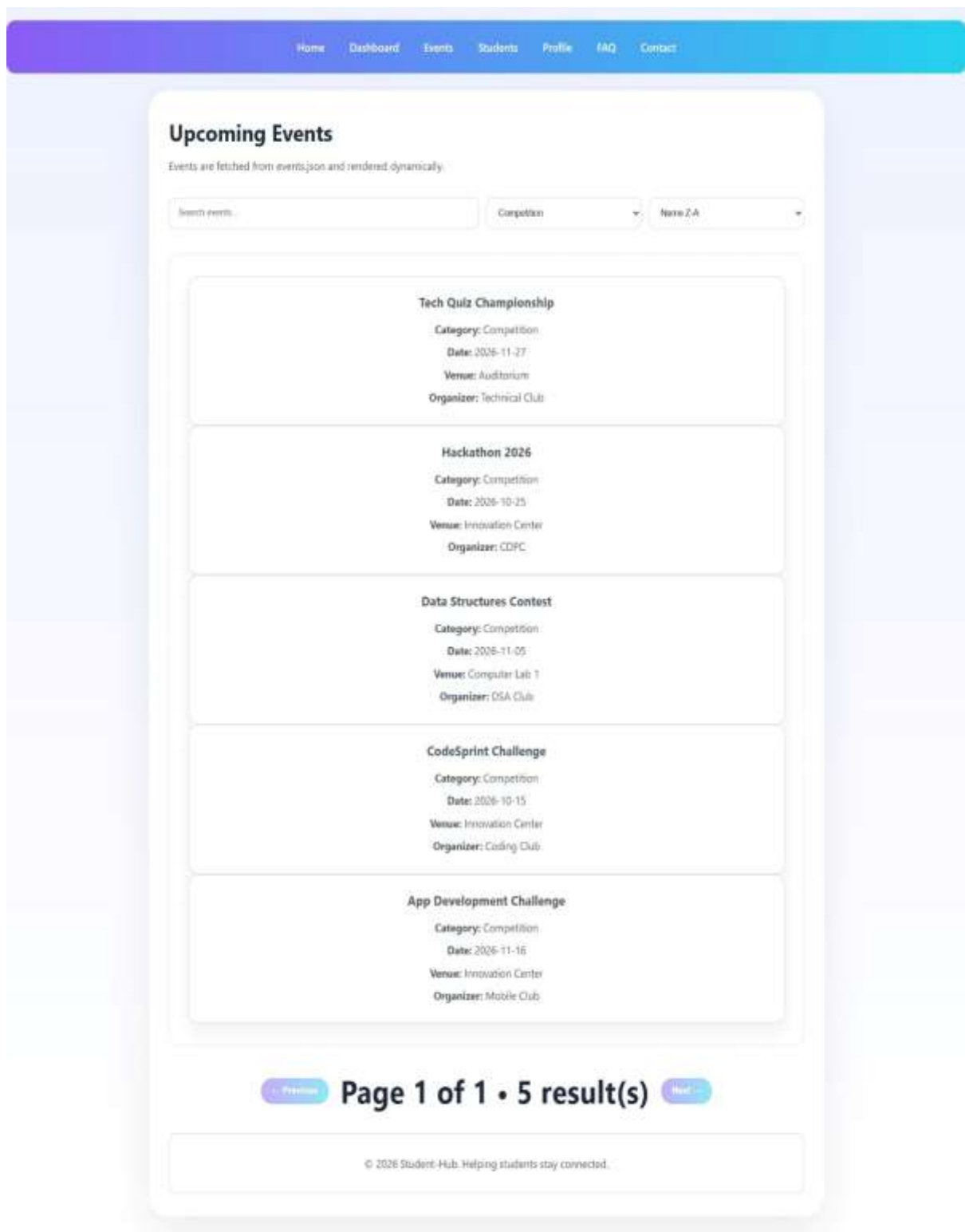
Next is a 'Quick Overview' section with the following statistics:

- Pending Assignments: 2
- Overall Attendance: 87%
- Upcoming Events: 3
- Unread Notifications: 4

Below the overview is a 'Quick Links' section with the following links:

- [View Assignments](#)
- [View Attendance](#)
- [View Events](#)
- [View Profile](#)

At the bottom of the page, there is a footer: © 2026 Student Hub. Helping students stay connected.

Event Page :

Contact page :

Contact Us

College: Student Hub University

Email: support@studenthub.ac.in

Phone: +91 98765 43210

Address: Student Hub Campus, India

[Send a Message](#)

Name:

Email:

Subject:

Message:

[Send Message](#)

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Profile page :

My Profile

Full Name:

Email:

Phone:

Course:

Semester:

Address:

[Update Profile](#)

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Register page :

Student-Hub
A polished student portal for assignments, attendance, events, and profile management.

Home Dashboard Assignments Attendance Events Profile FAQ Contact Login

Student Registration

Create your Student-Hub account by entering the required information.

Full Name *
Enter your full name

Email Address *
example@stuhub.com

Student ID *
Example: 202118

Mobile Number *
Enter 10 digit mobile number

Course *
Select Course

Year *
Select Year

Gender *
☐ Male
 ☐ Female
 ☐ Other

Password *
Create a strong password

Use at least 8 characters with uppercase, lowercase, number and special character.

Confirm Password *
Re-enter your password

☐ I agree to the terms and conditions *

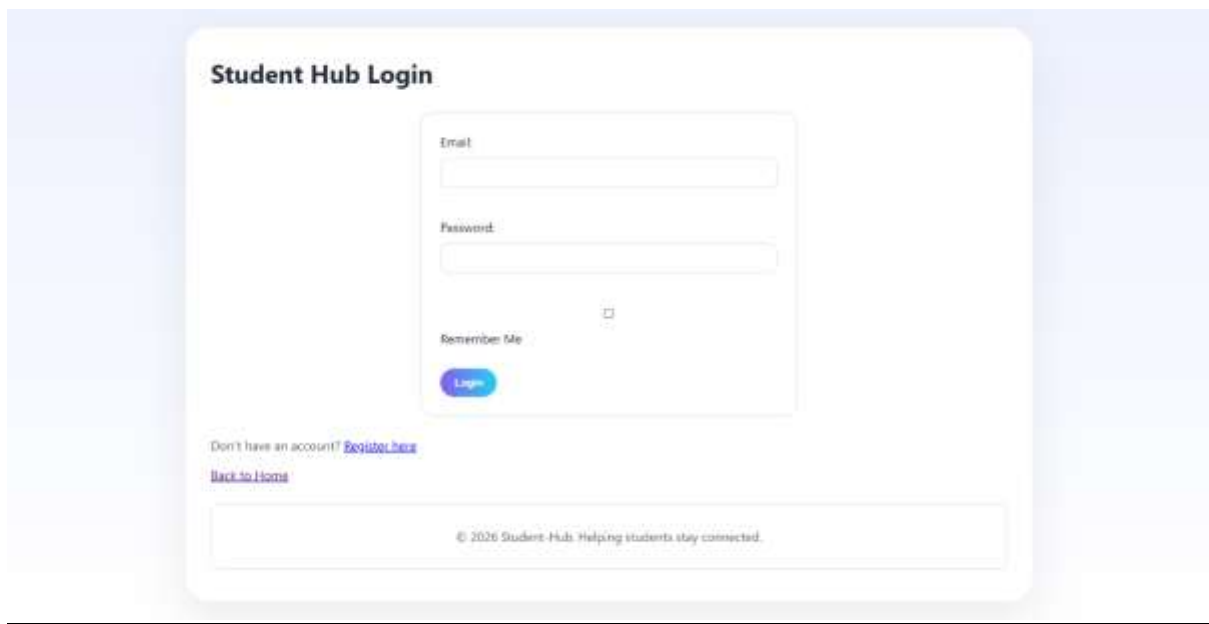
[Register](#)

Already have an account? [Login here](#)

[Back to Home](#)

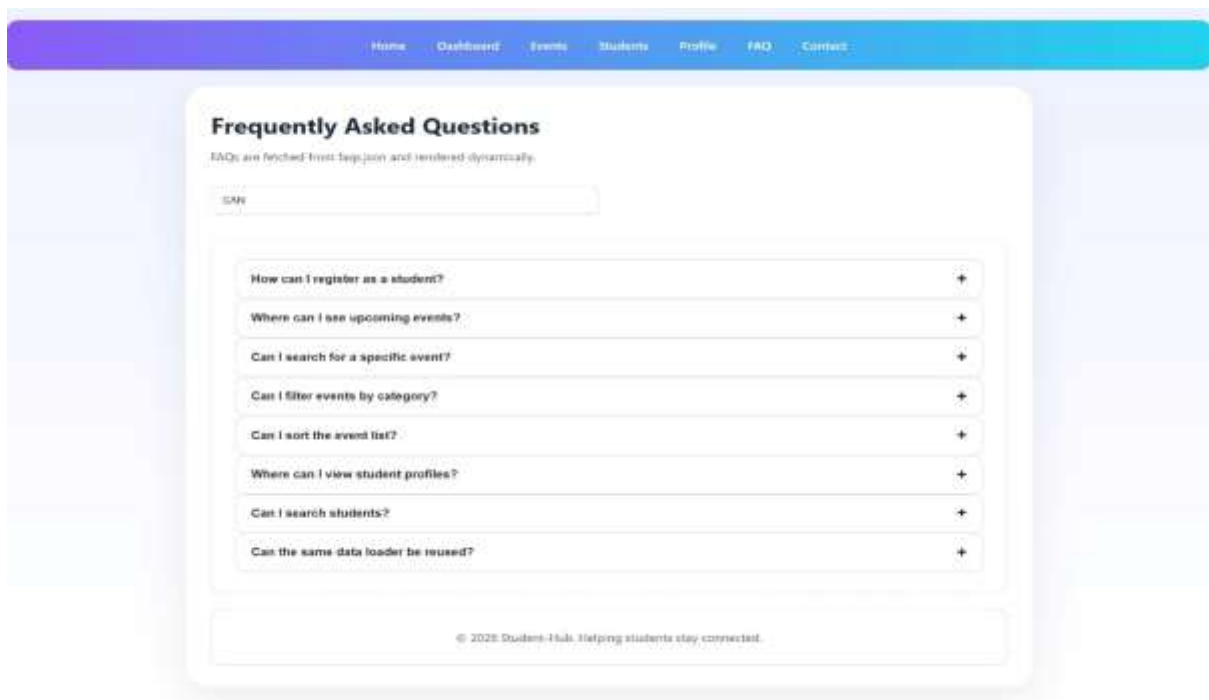
© 2020 Student-Hub. Helping students stay connected.

Login page :



The login page features a central white card with a light blue border. At the top of the card is the title "Student Hub Login". Below the title is a form with two input fields: "Email" and "Password". To the right of the "Password" field is a checkbox labeled "Remember Me". Below the form is a blue "Login" button. Underneath the button, there is a link "Don't have an account? Register here" and another link "Back to Home". At the bottom of the card, a footer states "© 2026 Student-Hub. Helping students stay connected."

Faqs page :



The FAQ page has a blue header with navigation links: Home, Dashboard, Events, Students, Profile, FAQ, and Contact. The main content area is a white card titled "Frequently Asked Questions". Below the title is a subtext: "FAQs are fetched from faqs.json and rendered dynamically." There is a search bar labeled "Search". Below the search bar is a list of eight FAQ items, each with a plus icon on the right: "How can I register as a student?", "Where can I see upcoming events?", "Can I search for a specific event?", "Can I filter events by category?", "Can I sort the event list?", "Where can I view student profiles?", "Can I search students?", and "Can the same data loader be reused?". At the bottom of the card, a footer states "© 2026 Student-Hub. Helping students stay connected."

Question & Answer:

1. How is JSON fetched, parsed, and rendered?

JSON data is fetched from external .json files using the Fetch API in JavaScript. The fetch() function sends a request to the JSON file and receives the response. The response is converted into JavaScript objects using response.json(). The data is then processed and dynamically rendered into HTML elements. This allows the webpage to display updated data without manually writing every record in HTML.

2. Which array methods are used for search, filter, sort, and pagination?

The filter() method is used to find records matching the search or selected filter. The sort() method is used to arrange records alphabetically or by date/year. The map() or forEach() method is used to create HTML for each record. The slice() method is used for pagination by selecting records for the current page. These array methods make data processing simple and efficient.

3. How are loading and error states handled?

A loading message such as “Loading events...” is displayed while the JSON data is being fetched. The Fetch API checks whether the response is successful before processing the data. If the JSON file cannot be loaded, an error is caught using try-catch. A user-friendly error message is then displayed instead of leaving the page blank. This improves the reliability and user experience of the application.

4. How is modularity maintained in JavaScript files?

JavaScript functionality is divided into separate files according to different responsibilities. A reusable data-loader.js file handles fetching JSON data. Separate files such as events.js, students.js, and faqs.js manage their respective pages. ES6 modules are used with import and export to share reusable functions. This makes the code organized, reusable, easier to maintain, and easier to debug.

Conclusion / Learning Outcome:

In this practical, we learned how to fetch and use external JSON data with the Fetch API and dynamically render events, student profiles, and FAQs using JavaScript. We implemented search, filtering, sorting, and pagination using JavaScript array methods such as `filter()`, `sort()`, `map()`, and `slice()`. We also handled loading and error states to provide a better user experience. Modular JavaScript files were used to keep the code organized, reusable, and easy to maintain. Overall, this practical helped us understand dynamic data rendering and modern data-driven frontend development.

Githublink :- <https://github.com/d26it118-png/Student-Hub.git>